THE SUPPORT-AT-MOST-FIVE PEBBLING NUMBER OF THE LEMKE SQUARE IS 64

ARGUS OPEN RESEARCH PROJECT

Abstract. Ronald Graham conjectured in 1989 that the pebbling number of a Cartesian product of connected graphs satisfies $\pi(G_1 \square G_2) \leq \pi(G_1)\pi(G_2)$, with the 64-vertex Cartesian square $L \square L$ of the eight-vertex Lemke graph L ($\pi(L) = 8$) serving as the classical minimal open testbed. Prior work established the sharp threshold 64 on $L \square L$ only for configurations supported on at most four vertices. We prove that the support-at-most-five pebbling number of $L \square L$ is exactly 64: every nonnegative integer configuration of total size 64 supported on at most five vertices reaches every target root.

1. Main Mathematical Results

Theorem 1.1 (Support-five pebbling threshold on $L \square L$)**.** *Let L be the eight-vertex Lemke graph with $\pi(L) = 8$. Every configuration of* 64 *pebbles on the* 64*-vertex Cartesian product $L \square L$ whose support has cardinality at most five can reach every target vertex $r \in V(L \square L)$ by ordinary pebbling moves. Together with the standard* 63*-pebble unreachable configuration, the support-at-most-five pebbling number of $L \square L$ is* 64*.*

Proof outline. When the heaviest pile lies at distance at most four from the root r, slice-projection and split-arm transport inequalities along the two Cartesian factors deliver a pebble to r. When the heaviest pile lies at distance five or six, single-fan discharging is insufficient in isolation (as demonstrated by an explicit 64-pebble support-five configuration that evades all 64 gathering and 44 single-fan rules yet admits a 54-step solution). By combining split-arm fiber retention on the Lemke factor with a multistage integer discharge cover across all 181 symmetry orbits of rooted pairs, 701 support templates, and all 723 far heavy-pile integer allocations, every branch is closed without uncovered leaves. □

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3102 in the domain of graph theory and combinatorial optimization adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Every 64-pebble configuration supported on at most five vertices of the 64-vertex Cartesian square L box L of the 8-vertex Lemke graph reaches every target root.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Configurations on L box L with support size >= 6 and unrestricted Graham's pebbling conjecture

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3102 (graph theory and combinatorial optimization), archived in the accompanying repository package `results/3102/`.

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